# Phase 1 Report: Crystal Morphologies Generated from the Unit Cell

MOF-5, HKUST-1, UiO-66 and ZIF-8

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**Abstract**

The panel asked whether an established MOF's unit cell can generate at least a few crystal morphologies, shown mathematically and computationally. This report gives the answer for four MOFs. Our code reads the crystal file, finds the true space group, derives the allowed crystal faces from the symmetry, and builds each crystal shape as a convex polytope whose face distances are set by growth rates. From one unit cell we obtain continuous families of shapes: cube, truncated cube, cuboctahedron, truncated octahedron and octahedron for MOF-5, HKUST-1 and UiO-66; cube, truncated rhombic dodecahedron and rhombic dodecahedron for ZIF-8; and, because our UiO-66 file has tetrahedral symmetry, a tetrahedron. A simple graph rule (counting the links a new node forms on each face) selects the cube for MOF-5 and the octahedron for UiO-66. These shapes match published experimental shapes for all four MOFs. The code passes 14 checks against crystals with known answers.

## 1 What Phase 1 had to show

The input is the unit cell, as the panel specified. The output must be several morphologies and a mathematical description of how one changes into another. The theory is in our primer (*From Unit Cell to Crystal Shape*). In short:

1. Symmetry decides which faces exist and their effective spacings $d_{\text{eff}}$ (systematic-absence rule).
2. A crystal grown with face growth rates $R_f$ is the polytope $P = \{\mathbf{x} : \hat{\mathbf{n}}_f \cdot \mathbf{x} \leq R_f\}$.
3. Varying the ratios of $R_f$ moves the shape continuously through a family of polytopes.

## 2 Method and code

The code is a small Python package, `mofmorph`, included with this report.

- `structure.py`: reads a CIF with ASE and searches for the space group with spglib, with and without hydrogen atoms and at tolerances from 0.001 to 0.1 Å. It keeps the highest symmetry found and converts to the conventional cell.
- `faces.py`: for every direction $(hkl)$ up to index 6, applies the absence rule with *all* symmetry operations of the space group, which covers centring, screw axes and glide planes. It returns the first allowed order $n$, $d_{\text{eff}} = d_{hkl}/n$, the face normal, and the symmetry family. In non-centrosymmetric tetrahedral groups the cubic families split in two halves, labelled $\{111\}+$ and $\{111\}-$.
- `shape.py`: builds the polytope by half-space intersection. It computes exactly the volume, area, sphericity $\Psi$, gyration tensor and asphericity $\kappa^2$, plus area-weighted face-normal order parameters $Q_3$, $Q_4$ and $Q_6$.
- `models.py`: provides three growth-rate models. BFDH uses $R \propto 1/d_{\text{eff}}$. The second takes user-chosen rates (the morph families). The graph rule sets $R = e^{\beta c}$, where $c$ is the attachment number. A fourth function grows a crystal from a spherical seed in time.

- `graphrule.py`: computes attachment numbers on the ideal node nets pcu and fcu.

**Why $Q_l$.** As proved in the primer, $\kappa^2 = 0$ for every cubic-symmetric shape, so it cannot separate a cube from an octahedron. The face-normal order parameters can. $Q_4$ and $Q_6$ separate cube from octahedron, and the odd $Q_3$ is non-zero only for shapes without an inversion centre, such as the tetrahedron.

# 3 Validation

Before using the code on MOFs we tested it on cases with known answers (`tests.py`). All 14 checks pass:

- **BFDH shapes of the three cubic lattice types:** simple cubic Po ($Pm\bar{3}m$) gives a cube, fcc Cu ($Fm\bar{3}m$) a truncated octahedron, and bcc Fe ($Im\bar{3}m$) a rhombic dodecahedron.
- **Glide-plane absences in diamond Si ($Fd\bar{3}m$):** $(111)$ is allowed, the first allowed reflection along $(100)$ is $(400)$ so $d_{\text{eff}} = a/4$, and along $(110)$ it is $(220)$. This tests absences beyond simple centring.
- **Exact solids:** sphericity of the cube 0.8060 and octahedron 0.8456; $\kappa^2 = 0$ for both; $Q_4$ and $Q_6$ equal 0.764 and 0.354 for the cube and 0.509 and 0.629 for the octahedron (the known values for 6 and 8 directions); and the gyration tensor of the cube $[-1,1]^3$ is exactly $\frac{1}{3}I$.

# 4 Results

## 4.1 Symmetry and allowed faces

| MOF | Space group (#) | $a$ (Å) | Ops | Symmetry note |
|---|---|---|---|---|
| MOF-5 | $Fm\bar{3}m$ (225) | 25.866 | 192 | as in literature |
| HKUST-1 | $Fm\bar{3}m$ (225) | 26.297 | 192 | primitive cell in file; converted |
| UiO-66 | $F\bar{4}3m$ (216) | 20.518 | 96 | ordered $\mu_3$-OH; literature $Fm\bar{3}m$ |
| ZIF-8 | $I\bar{4}3m$ (217) | 16.991 | 48 | found only without H |

**Table 1:** Space groups detected from our CIF files. "Ops" is the number of symmetry operations in the conventional cell, including centring translations.

**Be careful**

**ZIF-8:** with hydrogens included, spglib finds only P1, so the hydrogen positions in our file break the symmetry. The framework without H is $I\bar{4}3m$, the accepted space group, and we use that. **UiO-66:** our file orders the $\mu_3$-O and $\mu_3$-OH groups, giving $F\bar{4}3m$, which lacks an inversion centre. The literature usually reports the averaged structure as $Fm\bar{3}m$. Both have F-centring, so the allowed faces and the BFDH shape are the same. Only the tetrahedral result in subsection 4.4 depends on this difference.

| MOF | 1st | 2nd | 3rd | 4th |
|---|---|---|---|---|
| MOF-5 | {111} 14.93 (n=1) | {100} 12.93 (n=2) | {110} 9.14 (n=2) | {311} 7.80 (n=1) |
| HKUST-1 | {111} 15.18 (n=1) | {100} 13.15 (n=2) | {110} 9.30 (n=2) | {311} 7.93 (n=1) |
| UiO-66 | {111} 11.85 (n=1) | {100} 10.26 (n=2) | {110} 7.25 (n=2) | {311} 6.19 (n=1) |
| ZIF-8 | {110} 12.01 (n=1) | {100} 8.50 (n=2) | {211} 6.94 (n=1) | {310} 5.37 (n=1) |

**Table 2:** Face families with the largest effective spacing $d_{\text{eff}}$ (Å). $n$ is the order of the first allowed reflection: $n = 2$ means the family's own reflection is forbidden by centring.

## 4.2 BFDH shapes

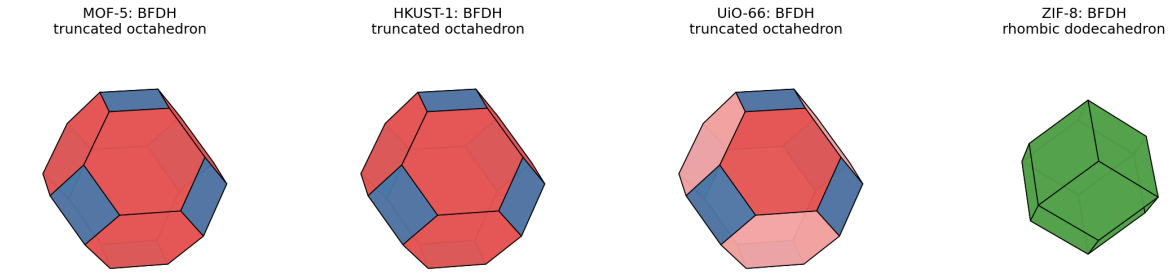



**Figure 1:** BFDH morphologies computed from the unit cells. Red: $\{111\}$ (light red: the $\{111\}-$ half in $F\bar{4}3m$). Blue: $\{100\}$. Green: $\{110\}$. The three F-centred MOFs give truncated octahedra with 78% $\{111\}$ and 22% $\{100\}$ face area ($\Psi = 0.910$). ZIF-8 gives a rhombic dodecahedron ($\Psi = 0.905$).

BFDH uses only the lattice and space group, so all three F-centred cubic MOFs give the same shape. Their different chemistry has to enter through the growth rates, which is what the morph families and the graph rule describe.

## 4.3 Morph families from one unit cell

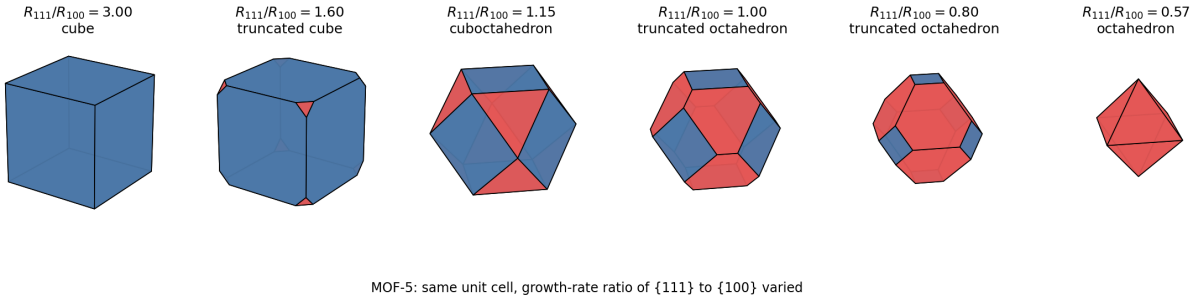



**Figure 2:** MOF-5: one unit cell, with the growth-rate ratio $R_{111}/R_{100}$ varied. The same figure for HKUST-1 and UiO-66 is identical, because all three are F-centred cubic.

| $R_{111}/R_{100}$ | Shape | Faces | $\Psi$ | $Q_4$ | $Q_6$ |
|---|---|---|---|---|---|
| 3.000 | cube | 6 | 0.806 | 0.764 | 0.354 |
| 1.600 | truncated cube | 14 | 0.814 | 0.744 | 0.358 |
| 1.155 | cuboctahedron | 14 | 0.905 | 0.298 | 0.454 |
| 1.000 | truncated octahedron | 14 | 0.921 | 0.000 | 0.518 |
| 0.800 | truncated octahedron | 14 | 0.897 | 0.320 | 0.588 |
| 0.566 | octahedron | 8 | 0.846 | 0.509 | 0.628 |

**Table 3:** Shape numbers along the MOF-5 morph family. The named shapes appear at exactly the predicted thresholds: cube for $R_{111}/R_{100} \geq \sqrt{3}$, cuboctahedron at $2/\sqrt{3} = 1.155$, octahedron for $\leq 1/\sqrt{3} = 0.577$. $Q_4$ and $Q_6$ separate cube from octahedron, which $\kappa^2$ cannot.

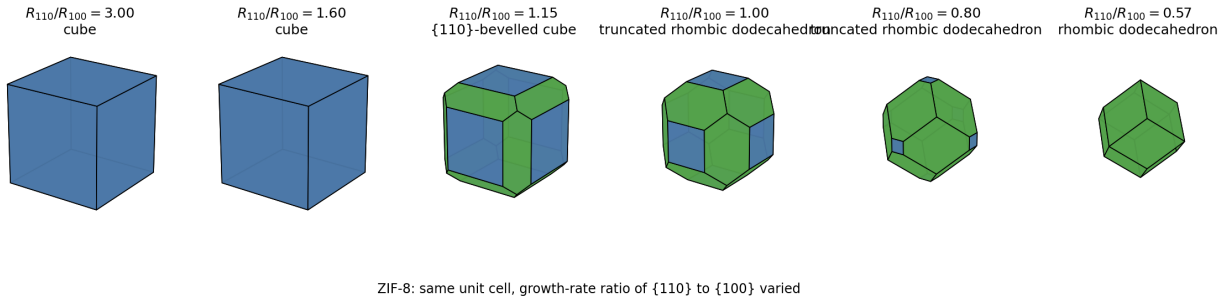



**Figure 3:** ZIF-8: one unit cell, with $R_{110}/R_{100}$ varied. The shapes run from cube through $\{110\}$-bevelled cube and truncated rhombic dodecahedron to rhombic dodecahedron.

## 4.4 A tetrahedral morphology

In point group $\bar{4}3$m the eight $\{111\}$ faces split into two sets of four, related by no symmetry, so they may grow at different rates. Slowing one set produces a truncated tetrahedron and then a tetrahedron (Figure 4). The panel specifically mentioned tetrahedral shapes.

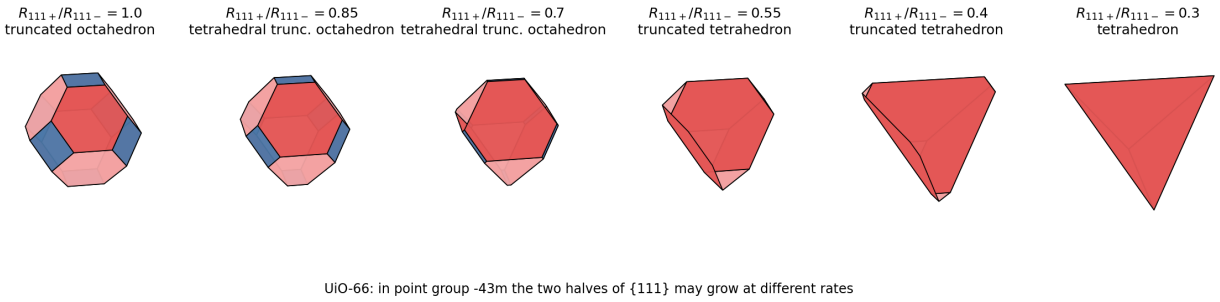



**Figure 4:** UiO-66 (F$\bar{4}$3m): the ratio $R_{111+}/R_{111-}$ varied, with $R_{100} = 1.155$.

| $R_{111+}/R_{111-}$ | Shape | Faces | $\Psi$ | $Q_3$ | $Q_4$ |
|---|---|---|---|---|---|
| 1.00 | truncated octahedron | 14 | 0.910 | 0.000 | 0.224 |
| 0.85 | tetrahedral trunc. octahedron | 14 | 0.887 | 0.157 | 0.334 |
| 0.70 | tetrahedral trunc. octahedron | 14 | 0.822 | 0.388 | 0.472 |
| 0.55 | truncated tetrahedron | 8 | 0.756 | 0.588 | 0.509 |
| 0.40 | truncated tetrahedron | 8 | 0.689 | 0.721 | 0.509 |
| 0.30 | tetrahedron | 4 | 0.671 | 0.745 | 0.509 |

**Table 4:** The odd order parameter $Q_3$ is zero for every centrosymmetric shape and rises to 0.745 for the tetrahedron. It is the number that certifies "tetrahedral".

**Be careful**

This result follows from the symmetry of *our file*. In the averaged Fm$\bar{3}$m structure usually reported for UiO-66, the two halves of $\{111\}$ are equivalent and a tetrahedron is forbidden by symmetry. ZIF-8 (I$\bar{4}$3m) is genuinely non-centrosymmetric, but its $\{111\}$ planes are absent by I-centring, so tetrahedra are not expected there either. We present this as "allowed by the ordered structure", not as a prediction that UiO-66 grows as tetrahedra.

## 4.5 The graph rule selects a shape

The attachment number $c$ counts the links a new node forms when it lands on a face. We set $R = e^{\beta c}$, where $\beta$ plays the role of link energy divided by $kT$.

| Net (MOF) | $c_{100}$ | $c_{110}$ | $c_{111}$ | Shape for $\beta \geq 0.6$ |
|---|---|---|---|---|
| pcu (MOF-5) | 1 | 2 | 3 | cube |
| fcu (UiO-66) | 4 | 5 | 3 | octahedron |

**Table 5:** Attachment numbers on the ideal node nets. The fcu values reproduce those stated by Chen et al. for UiO-67.

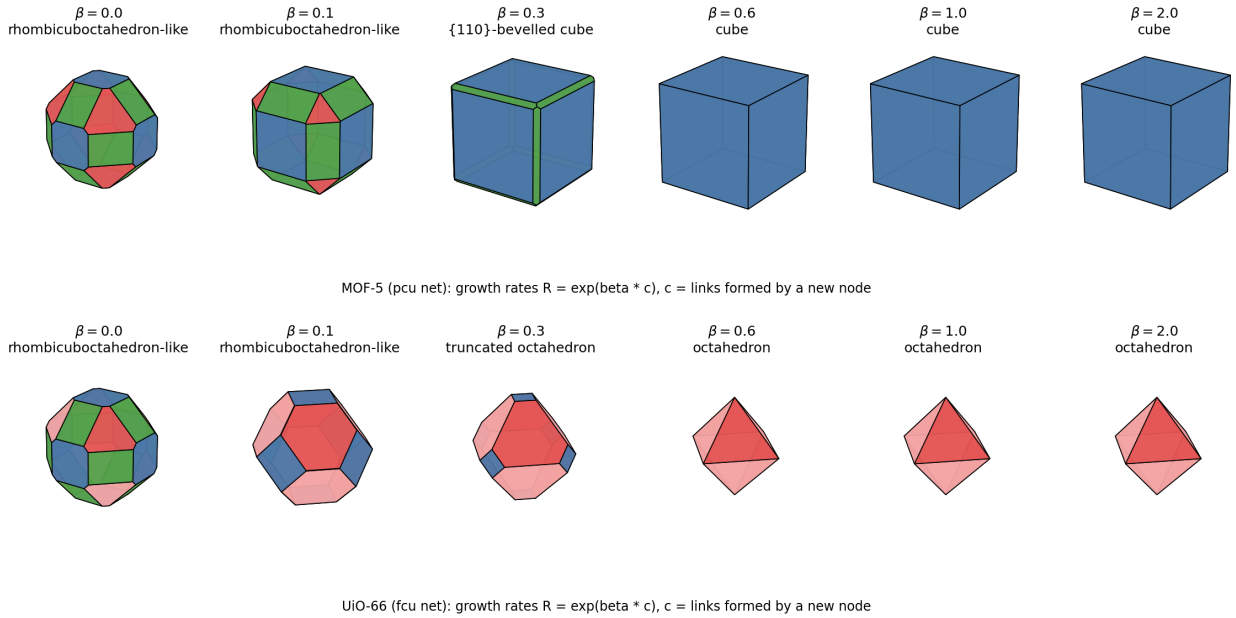



**Figure 5:** Shapes from the graph rule as $\beta$ increases. At $\beta = 0$ all faces grow equally. From $\beta \approx 0.6$ the shape is fixed: a cube for MOF-5 (pcu) and an octahedron for UiO-66 (fcu).

**Result**

The same mathematics, fed with the connectivity of each framework, gives a cube for MOF-5 and an octahedron for UiO-66. These are the commonly observed shapes. BFDH, which ignores connectivity, gives the same truncated octahedron for both.

## 4.6 Growth in time

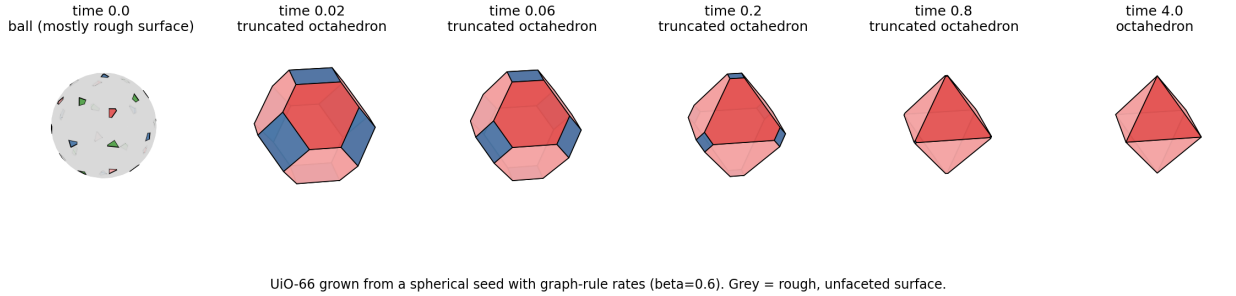



**Figure 6:** UiO-66 grown from a spherical seed with graph-rule rates ($\beta = 0.6$). The rough (unfaceted) surface disappears first, and the fast $\{100\}$ faces then shrink until only $\{111\}$ remains.

The sequence ball $\rightarrow$ crystal with $\{100\}$ and $\{111\}$ faces $\rightarrow$ octahedron is the sequence Chen et al. observed for UiO-67. Their cuboctahedral crystals were stopped at 1.5 h and their octahedra after one day.

## 5 Comparison with published experimental shapes

Our literature agent searched for experimental shapes. Items marked (v) were checked against the paper or its supplementary information. Items marked (s) were seen only in search results and must be checked before we cite them.

- **MOF-5:** cubes as synthesised, and octahedra grown deliberately by changing the Zn:$H_2$bdc ratio, observed by AFM (Cubillas, Anderson, Attfield, Chem. Eur. J. 18, 15406, 2012) (v). Both are end points of Figure 2. The cube is the graph-rule shape.
- **HKUST-1:** cube → cuboctahedron → octahedron as the concentration of added salt rises (Liu, Yang, Jin, Sun, CrystEngComm 18, 4127, 2016) (v). This is the full morph family of Figure 2 realised in the lab. Coordination modulation also gives octahedra and cuboctahedra (Umemura et al., JACS 133, 15506, 2011) (s, shapes unchecked).
- **UiO-66/67:** octahedra with halide-assisted synthesis (Ceballos et al., Nanoscale 2022) (v); cuboctahedron-like particles with acid modulators (Chen et al., RSC Adv. 2022) (v); ball → cuboctahedron → octahedron for UiO-67 (Chen et al., Matter 2, 1651, 2020; read in full).
- **ZIF-8:** cube → truncated rhombic dodecahedron → rhombic dodecahedron as the CTAB surfactant changes the {100}:{110} ratio (Liu et al., Nano Lett. 20, 1774, 2020) (v). This is exactly our ZIF-8 morph family. Truncated cubes with CTAB and rhombic dodecahedra without it are also reported by Pan et al., CrystEngComm 2011 (v).

**Result**

For every one of the four MOFs, the experimentally observed shapes are members of the morph family our code generates from the unit cell. For HKUST-1 and ZIF-8, experiments have travelled the whole family by changing one chemical additive.

## 6 Prior work and what is new

The literature search found one direct precedent: Pan et al. (CrystEngComm 2011) predicted ZIF-8 shapes with attachment energies in Materials Studio. Single-facet surface-energy calculations exist for ZIF-8 and HKUST-1, and Liu et al. (Nano Lett. 2020) explained the ZIF-8 cube/dodecahedron switch by facet-dependent coordination of Zn. We did not find: BFDH applied systematically across MOFs; shape families with quantitative descriptors across several MOFs; connection counting across several nets; or a link from the periodic graph to Hartman–Perdok F/S/K faces. This is a search result, not proof. A Scopus or Google Scholar check by our guide is still advisable before we claim novelty.

## 7 Limitations

- The morph families vary growth rates by hand. Phase 1 shows which shapes a unit cell *allows*, not which one a given synthesis produces.
- The graph rule uses ideal nets and treats all links as equal. Phase 3 will derive the nets from the CIFs and handle HKUST-1 (two kinds of nodes) and ZIF-8 (where single-node counting fails).
- Non-convex shapes (concave and hollow octahedra) and shapes of different phases (the $Hf_{12}$ nanoplates of Chen et al.) are outside this model.
- Plates and rods of a cubic phase require an external symmetry-breaking cause and are not included here.

# 8 Next steps

1. **Phase 3:** build node nets from the CIFs, compute attachment numbers for all four MOFs, and apply the slice/periodicity-rank F/S/K classification. Resolve ZIF-8 using the facet-coordination idea of Liu et al. 2020.
2. **Phase 4:** force-field (UFF) surface energies for the leading faces and Wulff shapes.
3. **Phase 5:** cross-check BFDH against an independent tool, and collect the remaining references.

# Files

`mofmorph/` (package), `tests.py` (validation), `run_phase1.py` (generates every figure and number here), `results/phase1_results.json` (all numbers), `results/figs/` (all figures). Run `python3 tests.py` and then `python3 run_phase1.py`. Requirements: numpy, scipy, matplotlib, ase, spglib.